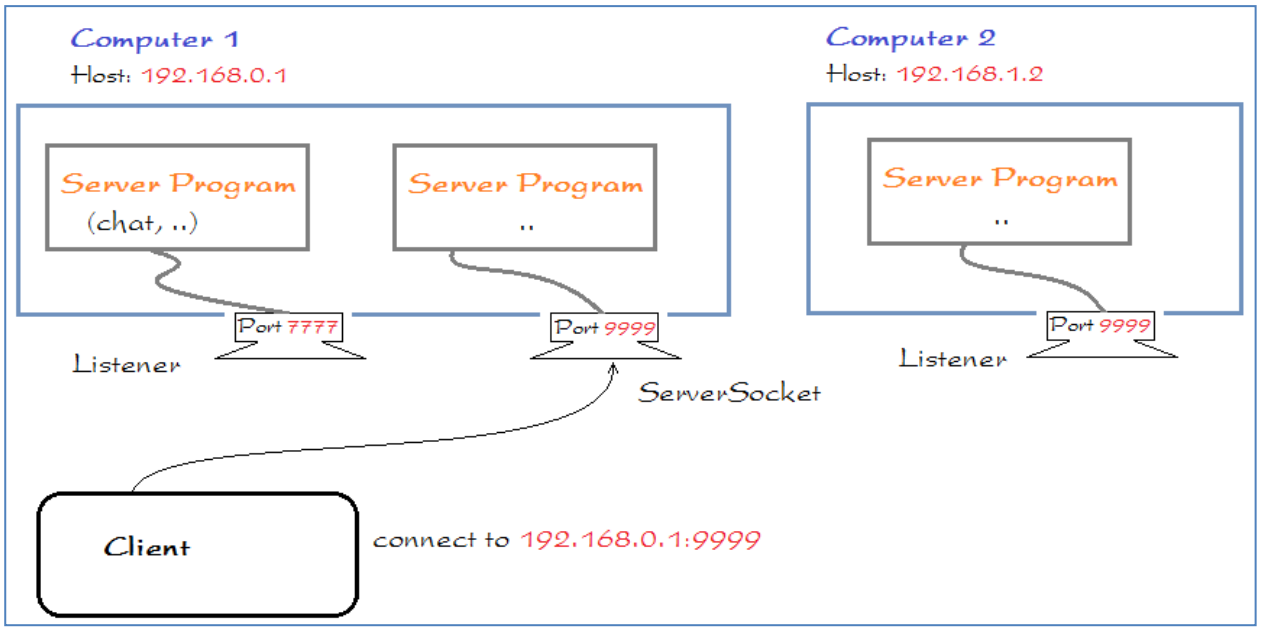
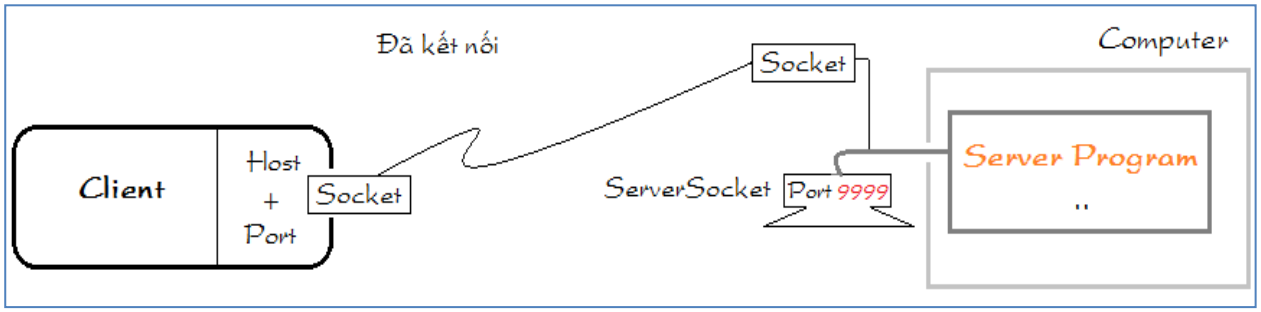
THỰC HÀNH BUỔI 3
(Socket-Java)





<u>Bài 1:</u> Mô hình TCP (server phục vụ tuần tự)
- Đổi các số từ 0 đến 9
Tạo file **EchoChatSingleServer.java**

```
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;
import java.net.ServerSocket;
import java.net.Socket;

public class EchoChatSingleServer {
   public final static int SERVER_PORT = 7;
   public static void main(String[] args) throws IOException {
      ServerSocket serverSocket = null;
      try {
         System.out.println("Đang kết nối đến " + SERVER_PORT + ", please wait  ...");
         serverSocket = new ServerSocket(SERVER_PORT);
         System.out.println("Server started: " + serverSocket);
         System.out.println("Waiting for a client ...");
         while (true) {
            try {
               Socket socket = serverSocket.accept();
```

```
                System.out.println("Client accepted: " + socket);

                OutputStream os = socket.getOutputStream();
                InputStream is = socket.getInputStream();
                int ch = 0;
                while (true) {
                    ch = is.read(); // Receive data from client
                    if (ch == -1) {
                        break;
                    }
                    os.write(ch); // Send the results to client
                }
                socket.close();
            } catch (IOException e) {
                System.err.println(" Connection Error: " + e);
            }
        }
    } catch (IOException e1) {
        e1.printStackTrace();
    } finally {
        if (serverSocket != null) {
            serverSocket.close();
        }
    }
  }
}
```

Tạo file **EchoChatClient.java**

```
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;
import java.net.Socket;
public class EchoChatClient {
  public final static String SERVER_IP = "127.0.0.1";
  public final static int SERVER_PORT = 7;
  public static void main(String[] args) throws IOException, InterruptedException {
    Socket socket = null;
    try {
      socket = new Socket(SERVER_IP, SERVER_PORT); // Connect to server
      System.out.println("Connected: " + socket);

      InputStream is = socket.getInputStream();
      OutputStream os = socket.getOutputStream();
      for (int i = '0'; i <= '9'; i++) {
        os.write(i); // Send each number to the server
        int ch = is.read(); // Waiting for results from server
        System.out.print((char) ch + " "); // Display the results received from the server
        Thread.sleep(200);
      }
    } catch (IOException ie) {
      System.out.println("Can't connect to server");
    } finally {
      if (socket != null) {
        socket.close();
      }
    }
  }
}
```

Bài 2: Mô hình TCP (server phục vụ tuần tự)
- Giải phương trình bậc nhất

Tạo file **Server_PTB1.java**

```
import java.io.*;
import java.net.*;
public class Server_PTB1 {
        public static void main(String[] args) throws IOException{
        ServerSocket ss = new ServerSocket(1234);
        System.out.println("Server da san sang");
        Socket cs = ss.accept();
        DataInputStream in = new DataInputStream(cs.getInputStream());
        DataOutputStream out = new DataOutputStream(cs.getOutputStream());
        int a,b;
        float x;
        a = in.readInt();
        b = in.readInt();
        //XU LY
        if(a==0)
          {
           if(b==0)
              out.writeUTF("X1");
           else
              out.writeUTF("X2");
          }
        else
          {
           out.writeUTF("X3");
           x=((float)(-b)/(float)a);
           out.writeFloat(x);
          }
        out.close();
        in.close();
        cs.close();
        ss.close();
    }
}
```

Tạo file **Client_PTB1.java**

```
import java.io.*;
import java.net.*;
import java.util.*;
public class Client_PTB1 {
    public static void main(String[] args) throws IOException{
        Socket cs = new Socket("localhost",1234);
        DataInputStream in = new DataInputStream(cs.getInputStream());
        DataOutputStream out = new DataOutputStream(cs.getOutputStream());
        int a,b;
        Scanner nhap = new Scanner(System.in);
        System.out.print("NNhập số a: ");
        a = nhap.nextInt();
        System.out.print("Nhập số b: ");
        b = nhap.nextInt();
        out.writeInt(a);
        out.writeInt(b);
        String k = in.readUTF();
        if(k.equalsIgnoreCase("X3"))
        {
```

```
        System.out.print("Nghiệm của phương trình là: ");
        float kq = in.readFloat();
        System.out.print(" x = "+kq);
      }
      if(k.equalsIgnoreCase("X2"))
        System.out.print("Phương trình vô nghiệm");

      if(k.equalsIgnoreCase("X1"))
        System.out.print("Phương trình vô số nghiệm");
      out.close();
      in.close();
      cs.close();
    }
}
```

Bài 3: Mô hình TCP (server phục vụ tuần tự)

Chat đơn giản

Tạo file **PC_Server.java**

```
import java.io.*;
import java.net.*;
public class PC_Server
{
  public static void main(String[] args) throws Exception
  {
     ServerSocket sersock = new ServerSocket(3000);
     System.out.println("Server  ready for chatting");
     Socket sock = sersock.accept( );

     // nhap lieu tu ban phim
     BufferedReader keyRead = new BufferedReader(new InputStreamReader(System.in));

     // goi chuoi ra outputstream su dung PrintWriter
     OutputStream ostream = sock.getOutputStream();
     PrintWriter pwrite = new PrintWriter(ostream, true);

     // doc chuoi cac bytes tu mot inputstream sư dung InputStreamReader va BufferedReader
     InputStream istream = sock.getInputStream();
     BufferedReader receiveRead = new BufferedReader(new InputStreamReader(istream));

     String receiveMessage, sendMessage;
     while(true)
     {
             //nhan message do client goi len socket
                 receiveMessage = receiveRead.readLine();
               if(receiveMessage != null)  // Kiểm tra xem có tin để nhận từ Client hay không
               {
                 System.out.print(">>Client: ");
                 System.out.println(receiveMessage);
               }

               //nhap du lieu tu ban phim
               System.out.print(">>Server: ");
               sendMessage = keyRead.readLine();

               //server goi du lieu len socket --> client
               pwrite.println(sendMessage);
               // pwrite.flush();
     }
```

```
    }
}
```

Tạo file **PC_Client.java**

```
import java.io.*;
import java.net.*;
public class PC_Client
{
  public static void main(String[] args) throws Exception
  {
    Socket sock = new Socket("localhost", 3000);
    // nhap du lieu tu ban phim
    BufferedReader keyRead = new BufferedReader(new InputStreamReader(System.in));
    // doc chuoi cac bytes tu mot inputstream su dung InputStreamReader va BufferedReader
    OutputStream ostream = sock.getOutputStream();
    PrintWriter pwrite = new PrintWriter(ostream, true);
    // goi chuoi ra outputstream su dung PrintWriter
    InputStream istream = sock.getInputStream();
    BufferedReader receiveRead = new BufferedReader(new InputStreamReader(istream));
    System.out.println("Bắt đầu chát, gõ nội dung từ bàn phím");
    String receiveMessage, sendMessage;
    while(true)
    {
          System.out.print("__Client: ");
          //nhap du lieu tu ban phim
          sendMessage = keyRead.readLine();
          //goi message len socket --> server
          pwrite.println(sendMessage);
          pwrite.flush();      // flush the data
          //nhan message do server goi len socket
          receiveMessage = receiveRead.readLine();
          if(receiveMessage != null) // Kiểm tra xem có tin để nhận từ server hay không
          {
         System.out.print("__Server: ");
         System.out.println(receiveMessage);

          }
     }
   }
}
```

<u>Bài 4:</u> Mô hình TCP (server phục vụ song song)

- Tính tổng 2 số:

Tạo file **TCPTong2soServer.java**

```
import java.io.*;
import java.net.*;

public class TCPTong2soServer {
        static int defaultPort = 7;
        public static void main(String[] args) throws IOException {
                ServerSocket ss = new ServerSocket(defaultPort);
                System.out.println("Server starting ....");
                while(true){
                        Socket s = ss.accept();
                        Tong2So thread = new Tong2So(s);
                         thread.start(); //cho chay luong
```

```
                }
        }
}
```

Tạo file **Tong2So.java**

```
import java.io.DataInputStream;
import java.io.DataOutputStream;
import java.io.IOException;
import java.net.Socket;

public class Tong2So extends Thread
{
        private Socket connectToClient;
        public Tong2So(Socket socket){
                connectToClient = socket;
        }
        public void run(){
         try{
                DataInputStream inFromClient = new DataInputStream(connectToClient.getInputStream());
                DataOutputStream outToClient = new
DataOutputStream(connectToClient.getOutputStream());
                while(true){
                        int so1 = inFromClient.readInt();
                        int so2 = inFromClient.readInt();
                        int tong = so1+ so2;
                        outToClient.writeInt(tong);
                }
                }catch(IOException ex){
                        System.err.println(ex);
                }
        }
}
```

Tạo file **TCPTong2soClient.java**

```
import java.io.*;
import java.net.*;
public class TCPTong2soClient {
        static int portServer = 7;
        static String ipServer = "localhost";
        public static void main(String arg[])
        {
                try {
                        Socket s = new Socket(ipServer, portServer);
                        DataInputStream inFromServer = new DataInputStream(s.getInputStream());
                        DataOutputStream outToServer = new DataOutputStream(s.getOutputStream());
                        BufferedReader banPhim = new BufferedReader(new
InputStreamReader(System.in));
                        while (true) {
                                System.out.print("Nhap so thu nhat: ");
                                int so1 = Integer.parseInt(banPhim.readLine());
                                System.out.print("Nhap so thu hai: ");
                                int so2 = Integer.parseInt(banPhim.readLine());
                                // gui du lieu len server
                                outToServer.writeInt(so1);
                                outToServer.writeInt(so2);
                                int tong = inFromServer.readInt();
                                System.out.println("Server: Tong hai so la  " + tong);
                        }
```

```
                } catch (IOException ex) {
                        System.err.println(ex);
                }
        }
}
```

Bài 5: Mô hình TCP (server phục vụ song song)

- Tìm số nguyên tố

Tạo file **SNT_Server.java**

```
import java.io.*;
import java.net.*;

public class SNT_Server {
        static int defaultPort = 7;
        public static void main(String[] args) throws IOException {
                // TODO Auto-generated method stub
                ServerSocket ss = new ServerSocket(defaultPort);
                System.out.println("Server starting ....");
                while(true){
                        Socket s = ss.accept();
                        Thread_TimSNT thread = new Thread_TimSNT(s);
                         thread.start(); //cho chay luong
                                }
                }
        }
```

Tạo file **Thread_TimSNT.java**

```
import java.io.DataInputStream;
import java.io.DataOutputStream;
import java.io.IOException;
import java.net.Socket;

//Đinh nghia lop ke thua tu Thread de xu lý mot ket noi moi
class Thread_TimSNT extends Thread
{
        private Socket connectToClient; //khoi tao mot socket de ket noi voi client
        public Thread_TimSNT(Socket socket)
        {
                connectToClient = socket;
        }
        // Cài dat hàm run() de thuc hien theo luong moi duoc tao ra
        public void run()
        {
         try{
                DataInputStream inFromClient = new
DataInputStream(connectToClient.getInputStream());
                DataOutputStream outToClient = new
DataOutputStream(connectToClient.getOutputStream());
                while(true)
                {
                        int so1 = inFromClient.readInt();
                        //int soNguyenTo;
                        for(int i = 1; i < so1; i++)
                        {
                                if(kiemTraSNT(i) == true)
                                {
                                        outToClient.writeInt(i);
```

```
                        }
                }
                }
        }catch(IOException ex){
                System.err.println(ex);
        }
    }
    public boolean kiemTraSNT(int so)
    {
        if(so <= 0)
            return false;
        else
        {
            for(int i = 2; i <= so/2; i++)
            {
                if(so % i == 0)
                    return false;
            }
        }
        return true;
    }
}
```

Tạo file **SNT_Client.java**

```
import java.io.*;
import java.net.*;

public class SNT_Client {
    static int portServer = 7;
    static String ipServer = "localhost";
    public static void main(String arg[]){
        try {
            Socket s = new Socket(ipServer, portServer);
            DataInputStream inFromServer = new DataInputStream(s.getInputStream());
            DataOutputStream outToServer = new DataOutputStream(s.getOutputStream());
            BufferedReader banPhim = new BufferedReader(new InputStreamReader(
                    System.in));

            System.out.print("Nhap 1 so: ");
            int num1 = Integer.parseInt(banPhim.readLine());

            // gui du lieu len server
            outToServer.writeInt(num1);
            System.out.print("Các số nguyên tố  < "+num1+" la ");
            while (true) {
                int num2 = inFromServer.readInt();
                System.out.print(" " + num2);
            }

        } catch (IOException ex) {
            System.err.println(ex);
        }
    }
}
```

Bài 6: Mô hình TCP (server phục vụ song song)

- Viết phương thức tính khoảng cách d của điểm A và điểm B theo công thức tính khoảng cách như sau: d = $\sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$

Tạo file **Client_TinhKhoangCach.java**

```
import java.io.*;
import java.net.*;
public class Client_TinhKhoangCach {
	static int portServer = 7;
	static String ipServer = "localhost";
	public static void main(String arg[])
	{
		try {
			Socket s = new Socket(ipServer, portServer);
			DataInputStream inFromServer = new DataInputStream(s.getInputStream());
			DataOutputStream outToServer = new DataOutputStream(s.getOutputStream());
			BufferedReader banPhim = new BufferedReader(new
InputStreamReader(System.in));
			while (true) {
				System.out.print("Nhap Xa: ");
				int xa = Integer.parseInt(banPhim.readLine());
				System.out.print("Nhap Ya: ");
				int ya = Integer.parseInt(banPhim.readLine());
				System.out.print("Nhap Xb: ");
				int xb = Integer.parseInt(banPhim.readLine());
				System.out.print("Nhap Yb: ");
				int yb = Integer.parseInt(banPhim.readLine());
				// gui du lieu len server
				outToServer.writeInt(xa);
				outToServer.writeInt(ya);
				outToServer.writeInt(xb);
				outToServer.writeInt(yb);
				double d = inFromServer.readDouble();
				System.out.println("Server: khoang cach d = " + d);
			}
		} catch (IOException ex) {
			System.err.println(ex);
		}
	}
}
```

Tạo file **Server_TinhKhoangCach.java**

```
import java.io.*;
import java.net.*;

public class Server_TinhKhoangCach {
	static int defaultPort = 7;
	public static void main(String[] args) throws IOException {
		ServerSocket ss = new ServerSocket(defaultPort);
		System.out.println("Server starting ....");
		while(true){
			Socket s = ss.accept();
			TinhKhoangCach thread = new TinhKhoangCach(s);
			 thread.start(); //cho chay luong
		}
	}
}
```

Tạo file **TinhKhoangCach.java**

```
import java.io.DataInputStream;
import java.io.DataOutputStream;
import java.io.IOException;
import java.net.Socket;

public class TinhKhoangCach extends Thread
{
	private Socket connectToClient;
	public TinhKhoangCach(Socket socket){
		connectToClient = socket;
	}
	public void run(){
	 try{
		DataInputStream inFromClient = new DataInputStream(connectToClient.getInputStream());
		DataOutputStream outToClient = new
DataOutputStream(connectToClient.getOutputStream());
		while(true){
			int xa = inFromClient.readInt();
			int ya = inFromClient.readInt();
			int xb = inFromClient.readInt();
			int yb = inFromClient.readInt();
			double d = Math.sqrt((xa-xb)*(xa-xb) + (ya-yb)*(ya-yb));
			outToClient.writeDouble(d);
		}
		}catch(IOException ex){
			System.err.println(ex);
		}
	}
}
```

<u>Bài 7:</u> Mô hình UDP

Chat

Tạo file **MyClient.java**

```
import java.net.*;
import java.io.*;
public class MyClient{
   public static void main(String args[])throws Exception{
   Socket s=new Socket("localhost",3333);
      DataInputStream din=new DataInputStream(s.getInputStream());
      DataOutputStream dout=new DataOutputStream(s.getOutputStream());
      BufferedReader br=new BufferedReader(new InputStreamReader(System.in));

      String str="",str2="";
      while(!str.equals("stop")){
         System.out.print("Client: ");
         str=br.readLine();
         dout.writeUTF(str);
         dout.flush();
         str2=din.readUTF();
         System.out.print("Server says: "+str2);
         System.out.println("");
      }
      dout.close();
      s.close();
   }
}
```

Tạo file **MySever.java**

```
package Buoi4_bai4_UDP_Tuantu_Chat;
import java.net.*;
import java.io.*;
public class MyServer{
   public static void main(String args[])throws Exception{
      ServerSocket ss=new ServerSocket(3333);
      Socket s=ss.accept();
      DataInputStream din=new DataInputStream(s.getInputStream());
      DataOutputStream dout=new DataOutputStream(s.getOutputStream());
      BufferedReader br=new BufferedReader(new InputStreamReader(System.in));

      String str="",str2="";
      while(!str.equals("stop")){
         str=din.readUTF();
         System.out.print("client says: "+str);
         System.out.println("");
         System.out.print("Server: ");
         str2=br.readLine();
         dout.writeUTF(str2);
         dout.flush();
      }
      din.close();
      s.close();
      ss.close();
   }
}
```

<u>Bài 8:</u> Mô hình UDP

- Tính số chẵn lẻ

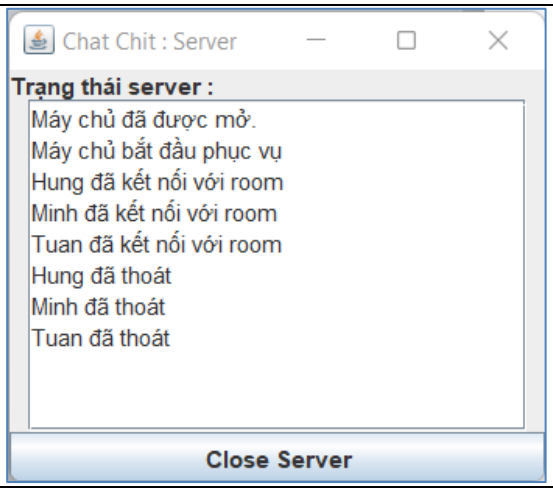
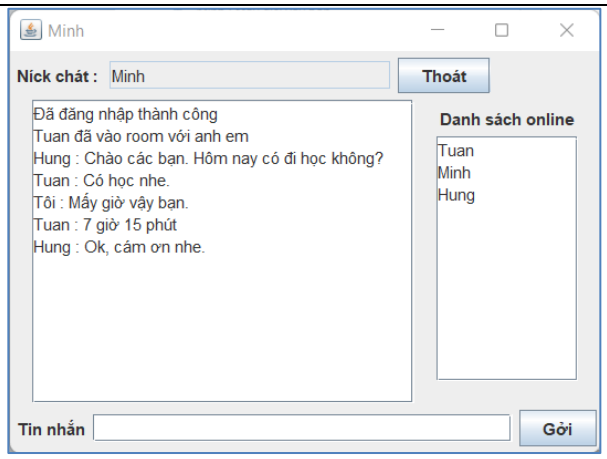
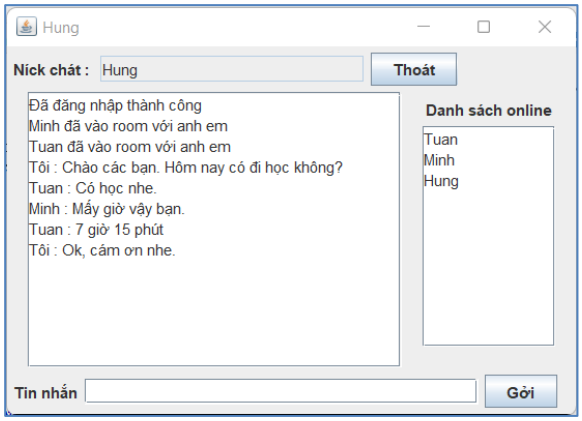
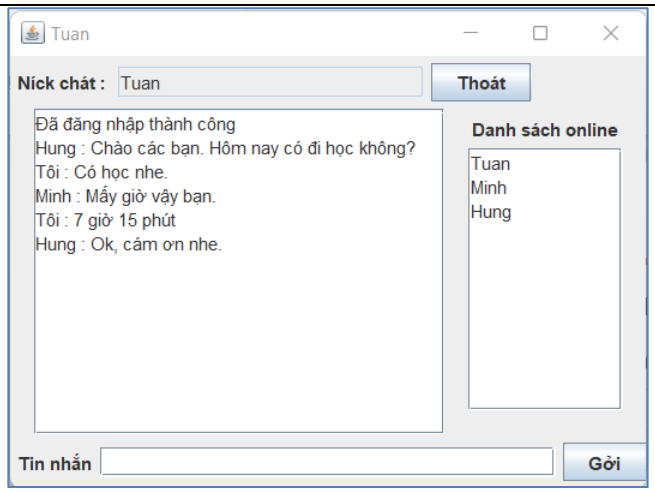
Tạo file **UDP_Server.java**

```
import java.io.*;
import java.net.*;

public class UDP_Server {
  public static void main(String[] args) throws IOException{
     DatagramSocket sv = new DatagramSocket(2000);
     // we expect a number less than 15 digits....
     byte buf[] = new byte[15];
     DatagramPacket q = new DatagramPacket(buf,buf.length);
     sv.receive(q);
     String kq = new String(q.getData()).trim();
     int m = Integer.parseInt(kq);
     byte buf2[] = ((m%2==0)? "Số chẵn":"Số lẻ").getBytes();
     InetAddress addcl = q.getAddress();
     int portcl=q.getPort();
     DatagramPacket k = new DatagramPacket(buf2,buf2.length,addcl,portcl);
     sv.send(k);
     sv.close();
  }
}
```

Tạo file **UDP_Client.java**

```
import java.io.*;
import java.net.*;
import java.util.*;

public class UDP_Client {
  public static void main(String[] args) throws IOException{
      DatagramSocket cl = new DatagramSocket();
      Scanner nhap = new Scanner(System.in);
      System.out.print("Nhập vào 1 số: ");
      int x = nhap.nextInt();
      // Ditect Hanoi-Saigon :o)
      byte buf[] = (x+"").getBytes();
      InetAddress addsv = InetAddress.getByName("localhost");
      int portsv = 2000;
      DatagramPacket p = new DatagramPacket(buf,buf.length,addsv,portsv);
      cl.send(p);
      byte buf2[]= new byte[256];
      DatagramPacket t = new DatagramPacket(buf2,buf2.length);
      cl.receive(t);
      String kq = new String(t.getData()).trim();
      System.out.print("kết quả " +kq);
      cl.close();
  }
}
```

BÀI TẬP LÀM THÊM

<u>Bài 9:</u> Mô hình TCP

- Viết chương trình chat nhiều máy tính như sau: